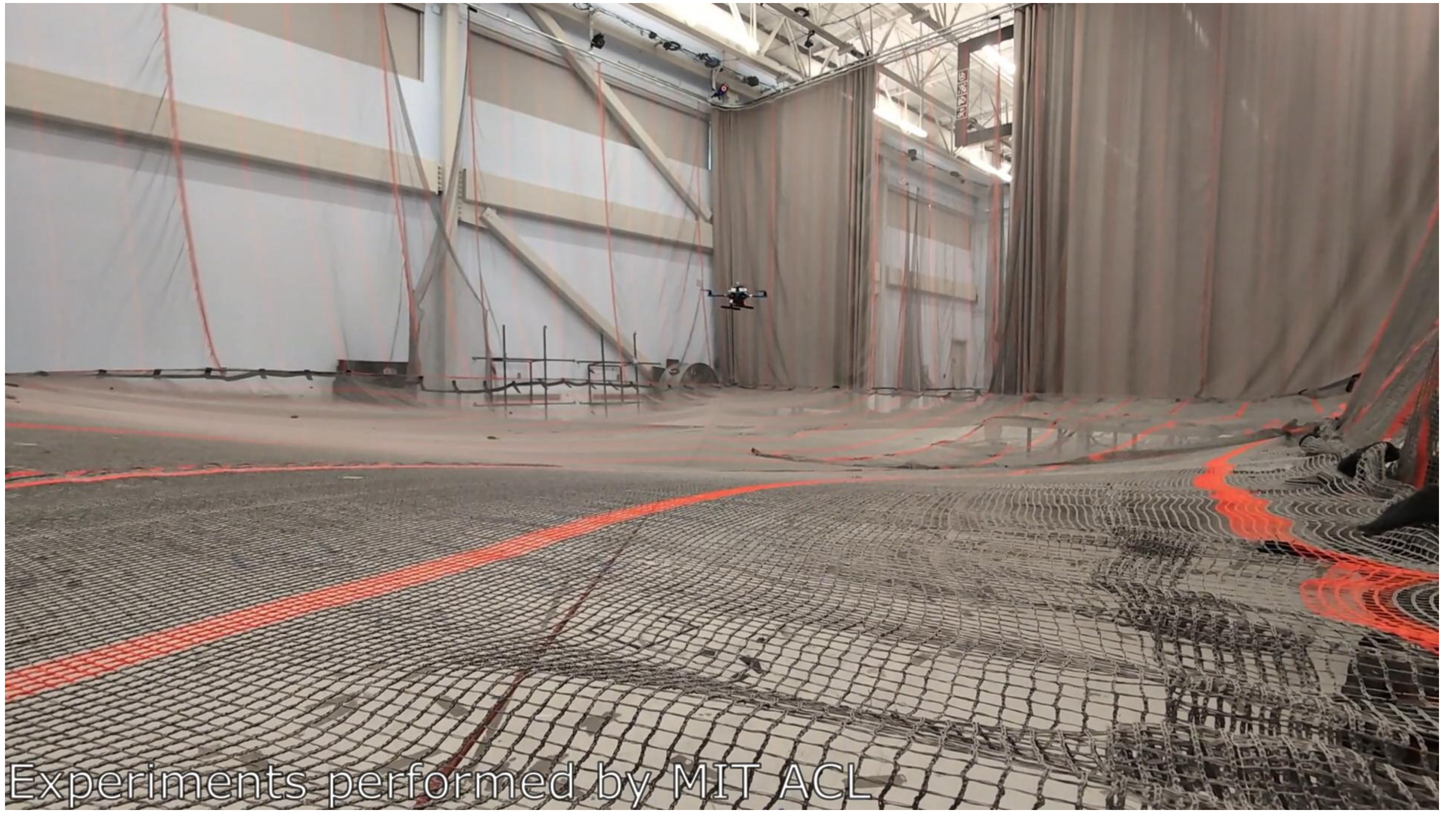
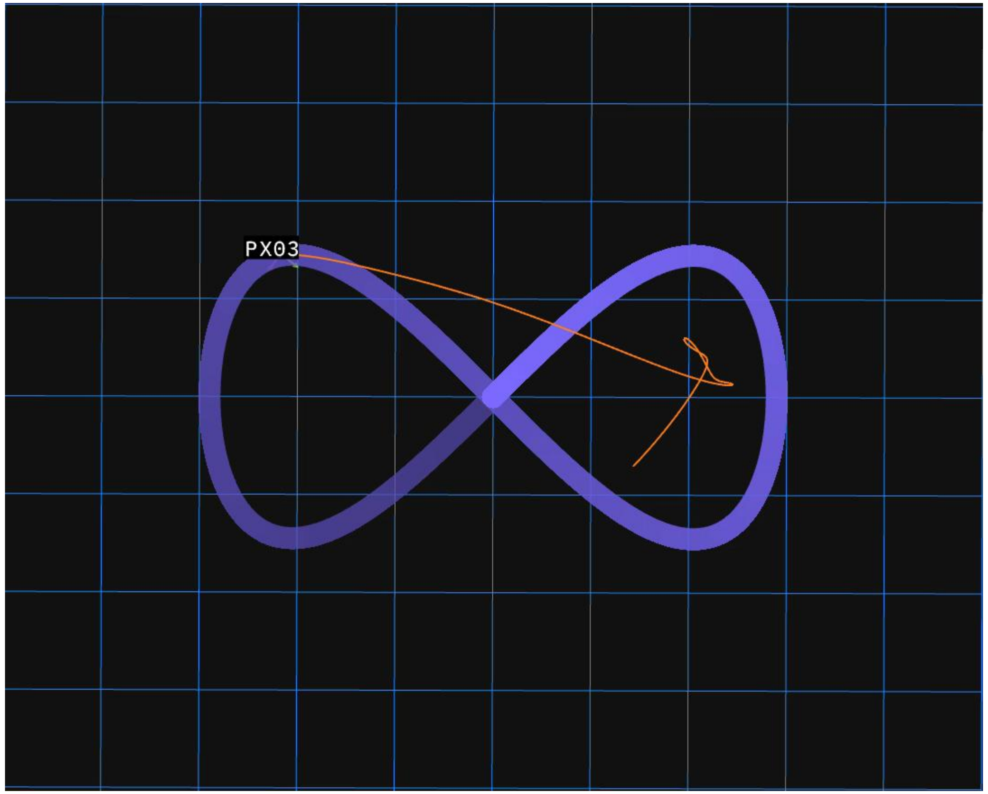
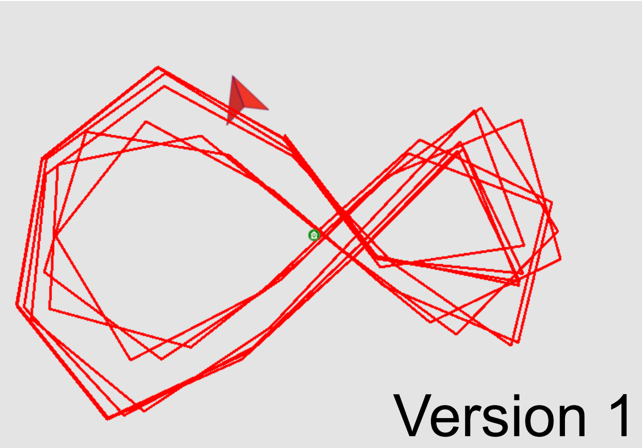
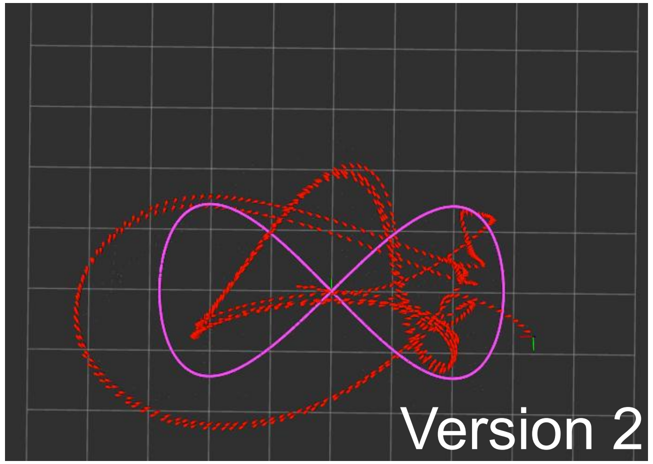
DEBUGGING PREVIOUS PX4 IMPLEMENTATION

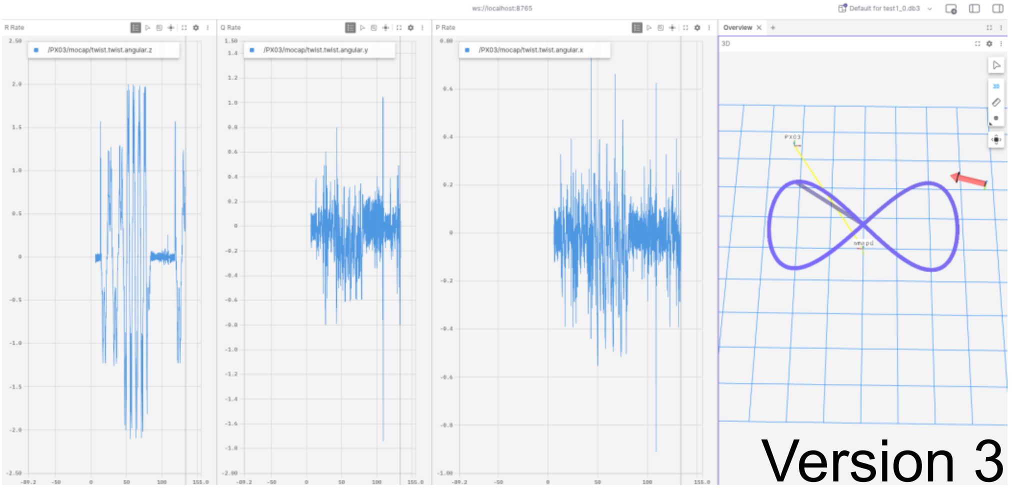


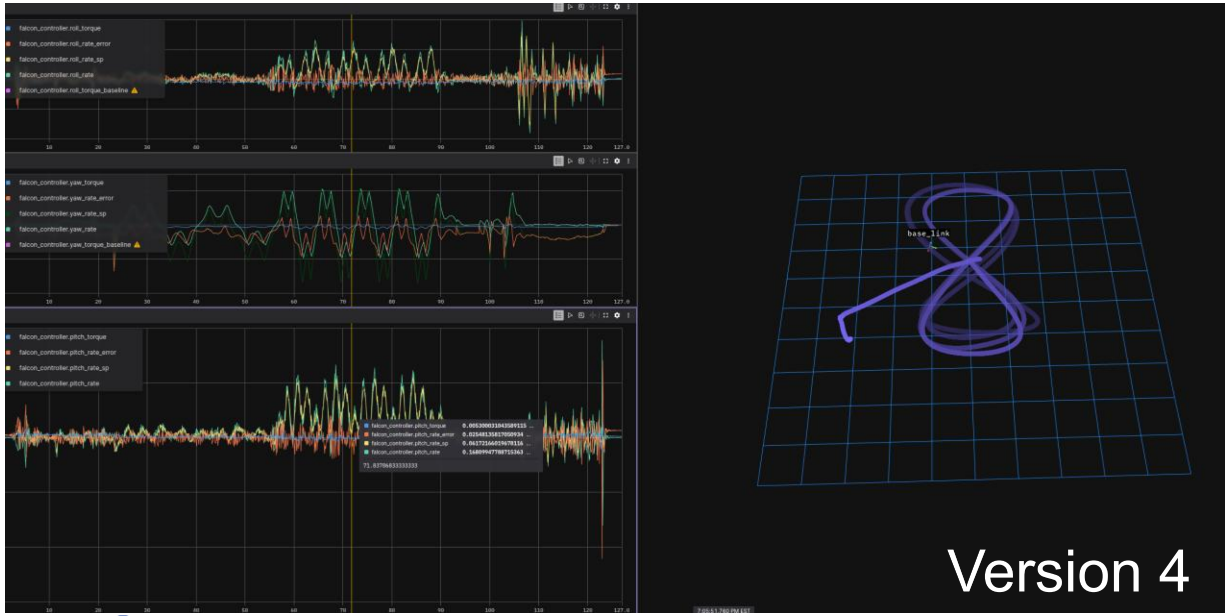


FOXGLOVE VISUALIZATION TOOLS









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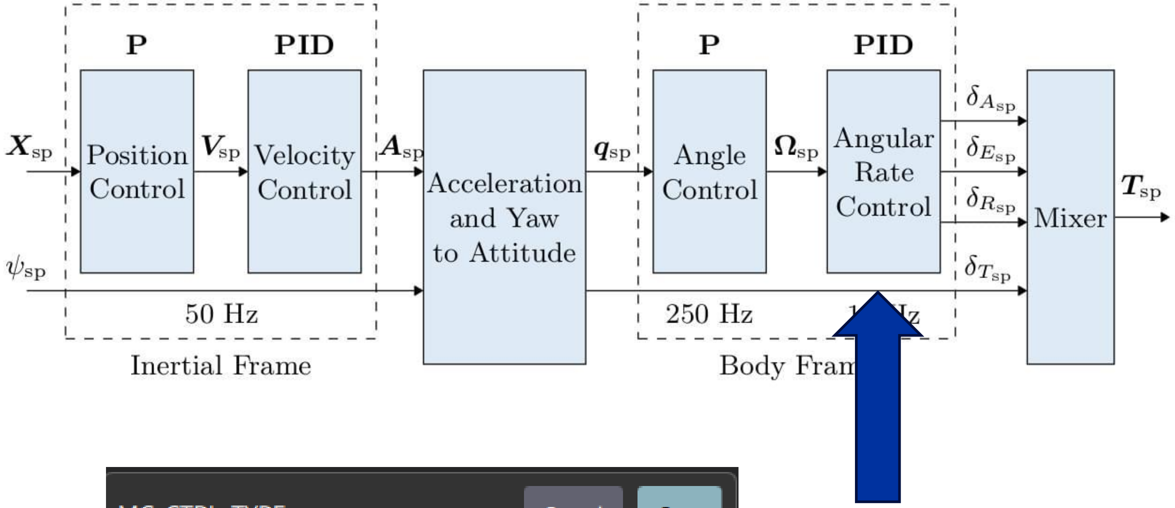
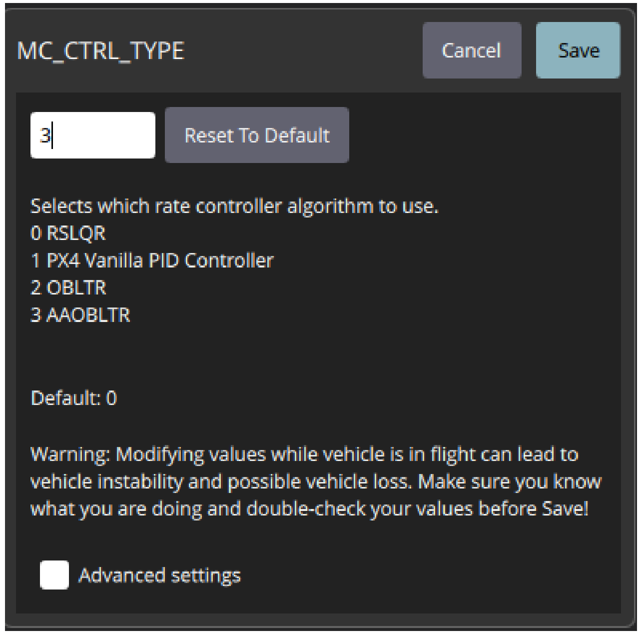
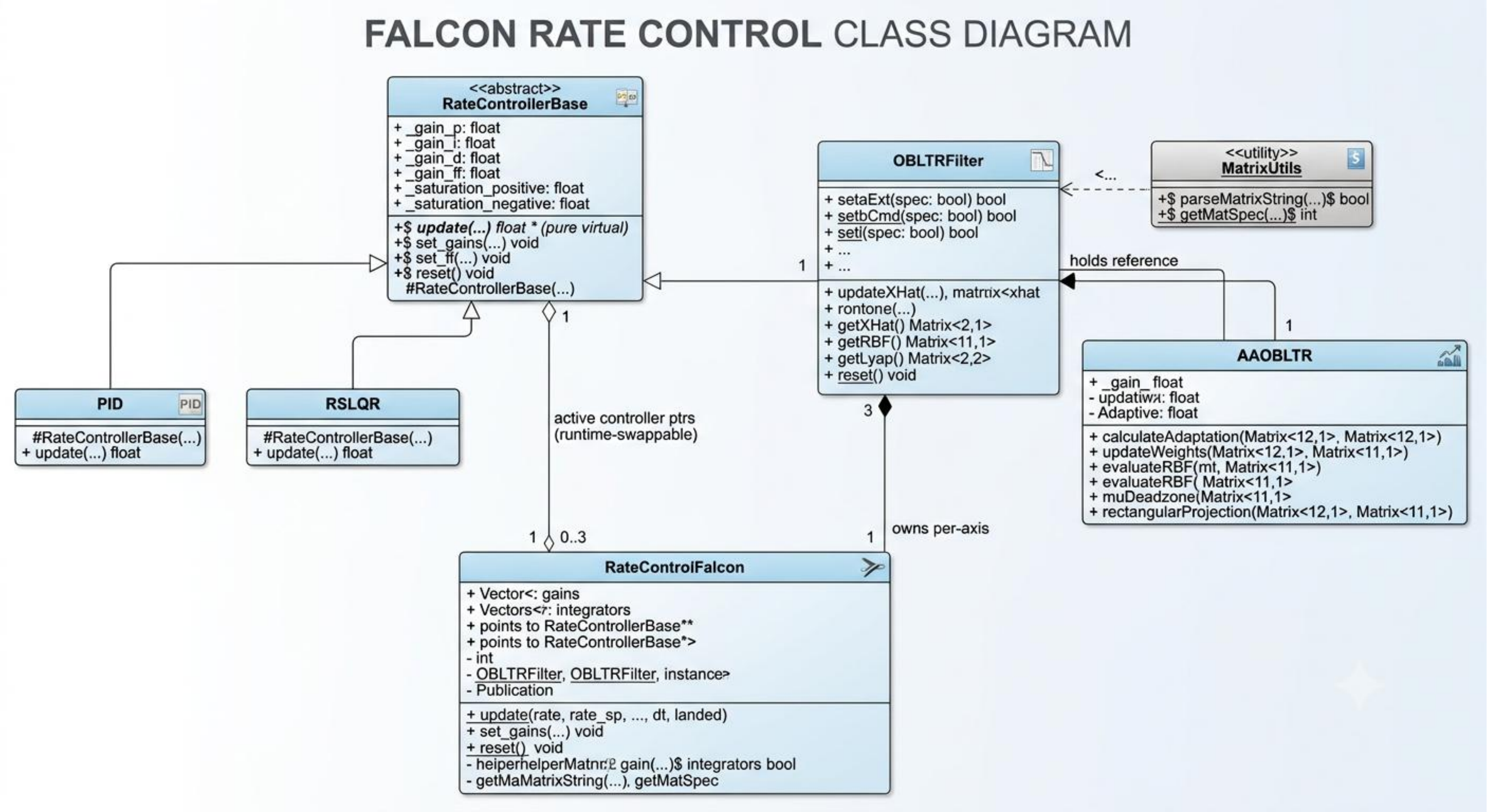
CUSTOM RATE CONTROLLER LIBRARY IN PX4

- For controller implementation
  - Working in the Angular Rate Control
- Forking and building custom PX4 Firmware
- Added new message types
  - Debugging
- Customized QGC to add custom fields





CUSTOM RATE CONTROLLER LIBRARY IN PX4



ADAPTIVE AUGMENTED OBSERVER-BASED CONTROL WITH LOOP TRANSFER RECOVERY (AAOBLTR)

- OBLTR:
  - A control method that estimates information that the drone can't directly senses, then uses it to fly as smoothly and reliably as if it could measure everything perfectly
- AAOBLTR:
  - An upgraded version that also adjusts itself on the fly, so the drone keeps flying well if conditions change

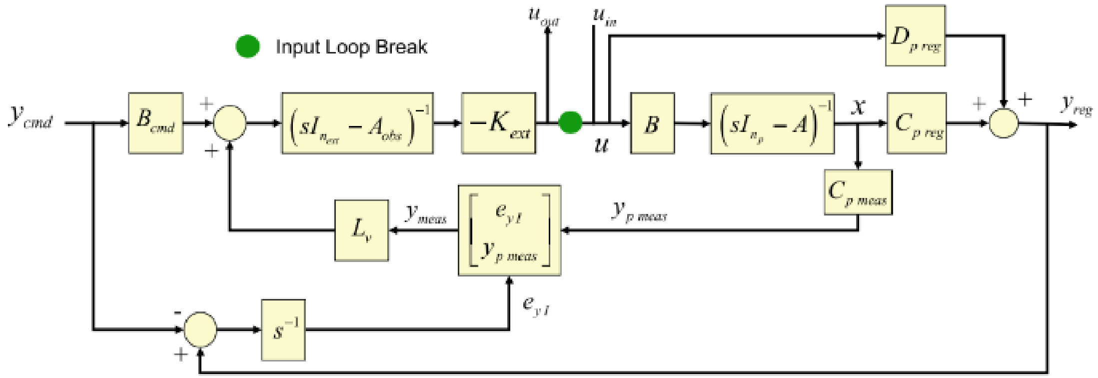
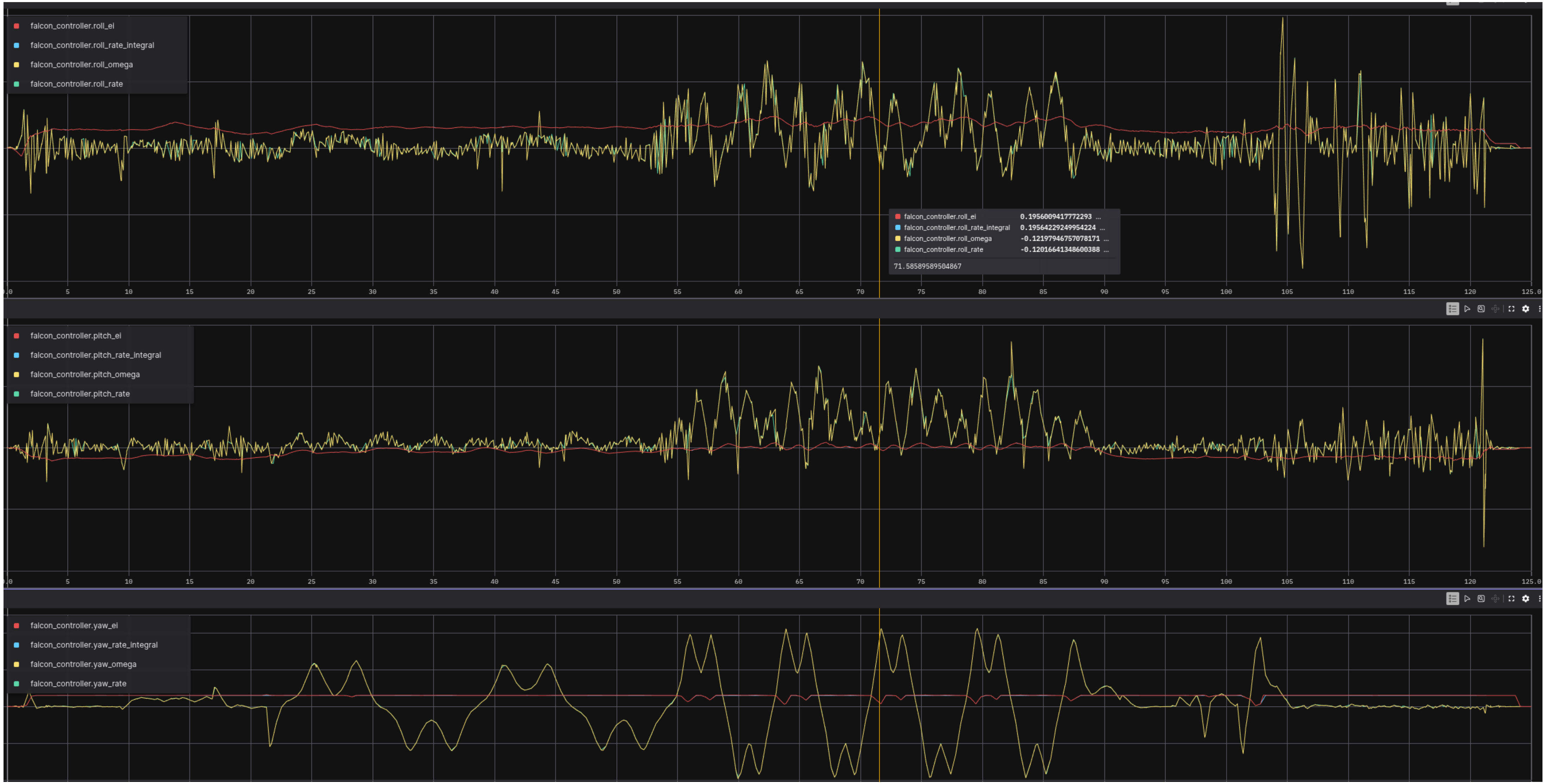


Fig. 6.13 Input loop gain calculation block diagram

TUNING OBLTR / AAOBLTR



- Ei == Rate Integral
- Omega == Pitch Rate

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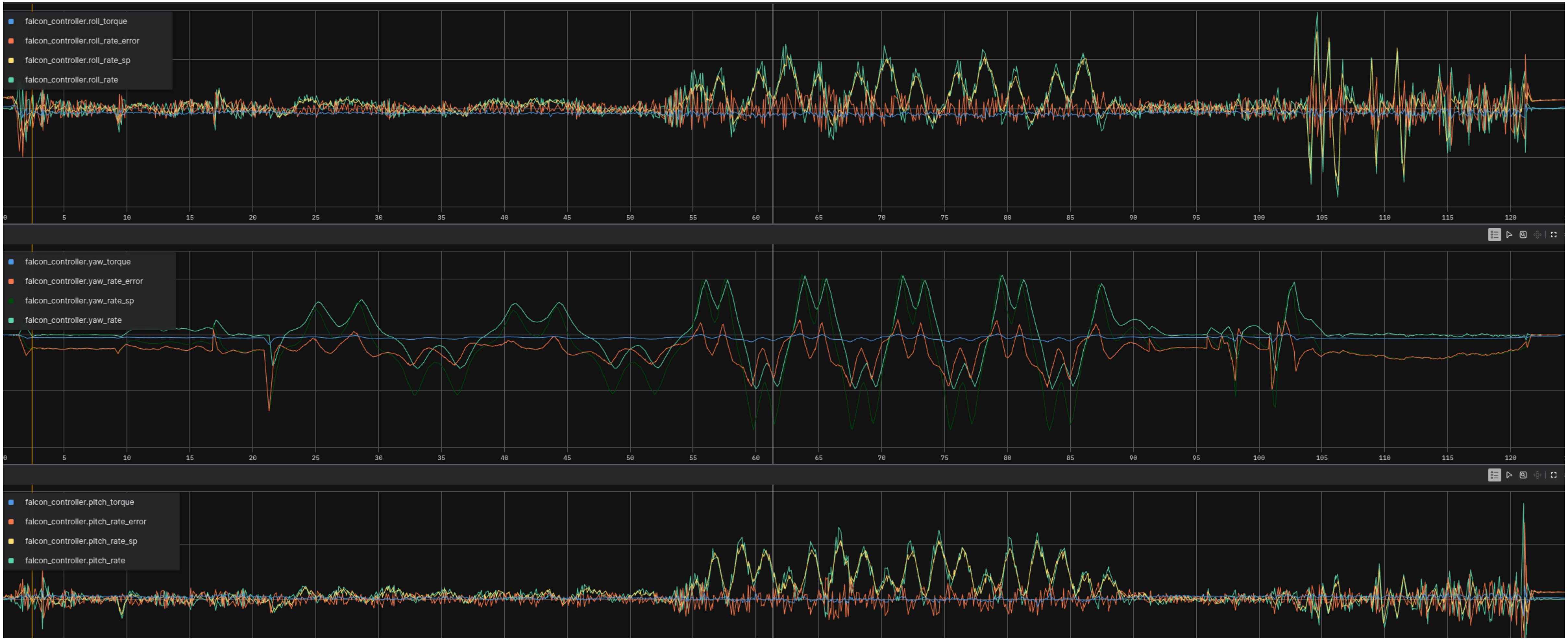
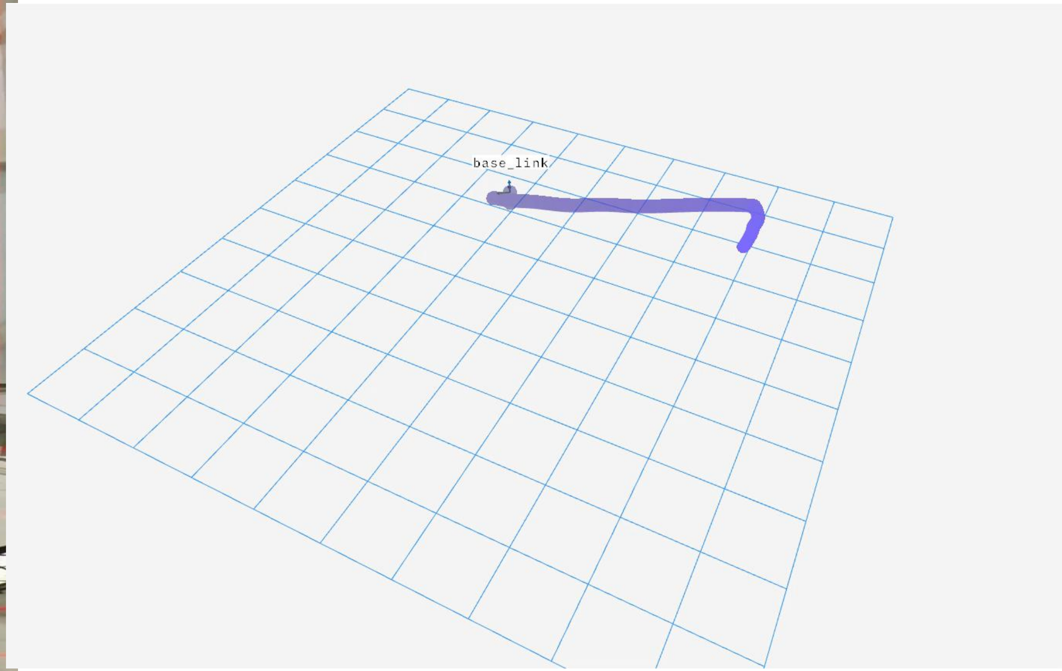
TUNING OBLTR / AAOBLTR



- Rates are tracking rate setpoints well

AAOBLTR DOING A FIGURE 8



OBSTACLE AVOIDANCE ANALYTICAL CBF SIMULATION

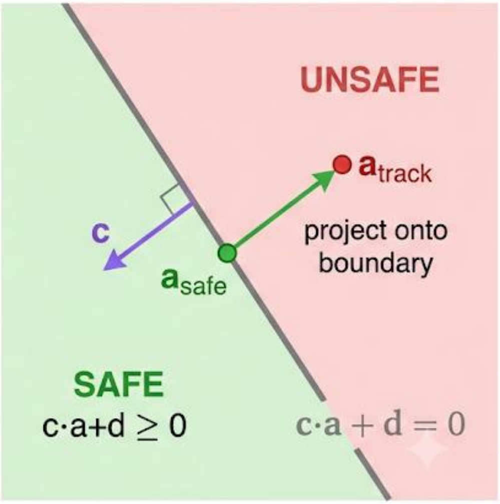
- Designed an Analytical CBF to avoid an obstacle set in the path of the Figure 8 mission

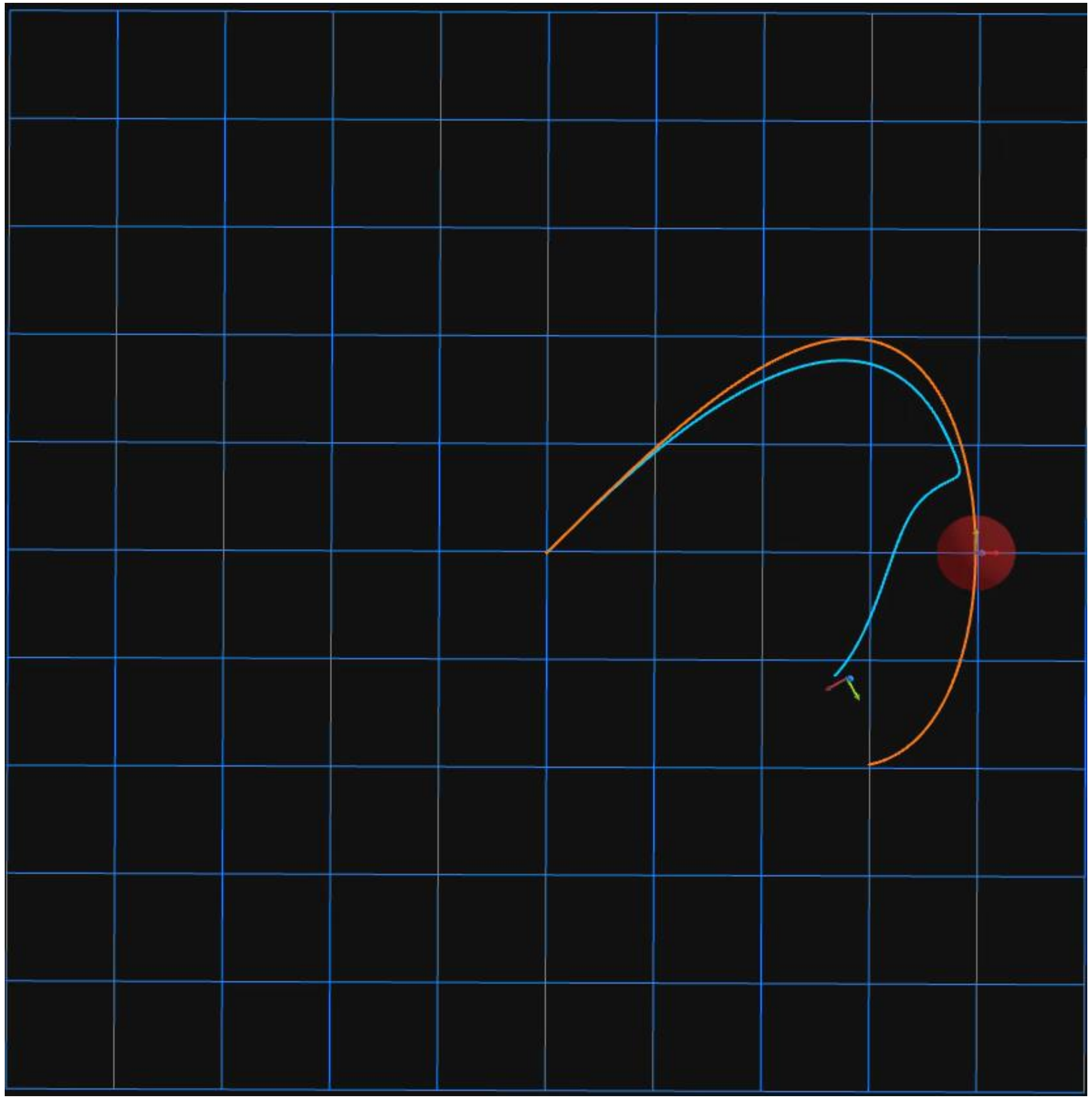
$$\Delta\mathbf{p} = \mathbf{p} - \mathbf{o}$$
$$h = \|\Delta\mathbf{p}\|^2 - r^2$$
$$\psi_1 = 2\,\Delta\mathbf{p}\cdot\mathbf{v} + \alpha_1 h$$
$$c = 2\Delta\mathbf{p}$$
$$d = 2\|\mathbf{v}\|^2 + 2\alpha_1(\Delta\mathbf{p}\cdot\mathbf{v}) + \alpha_2\psi_1$$
$$\text{margin} = c\cdot\mathbf{a}_{\text{track}} + d$$

Solution:

$$\mathbf{a}_{\text{safe}} = \mathbf{a}_{\text{track}},\ \text{margin} \geq 0$$
$$\mathbf{a}_{\text{safe}} = \mathbf{a}_{\text{track}} - c(\text{margin}/\|\mathbf{c}\|^2),\ \text{margin} < 0$$

One active obstacle ⇒ one linear inequality in a ⇒ closed form, no QP.





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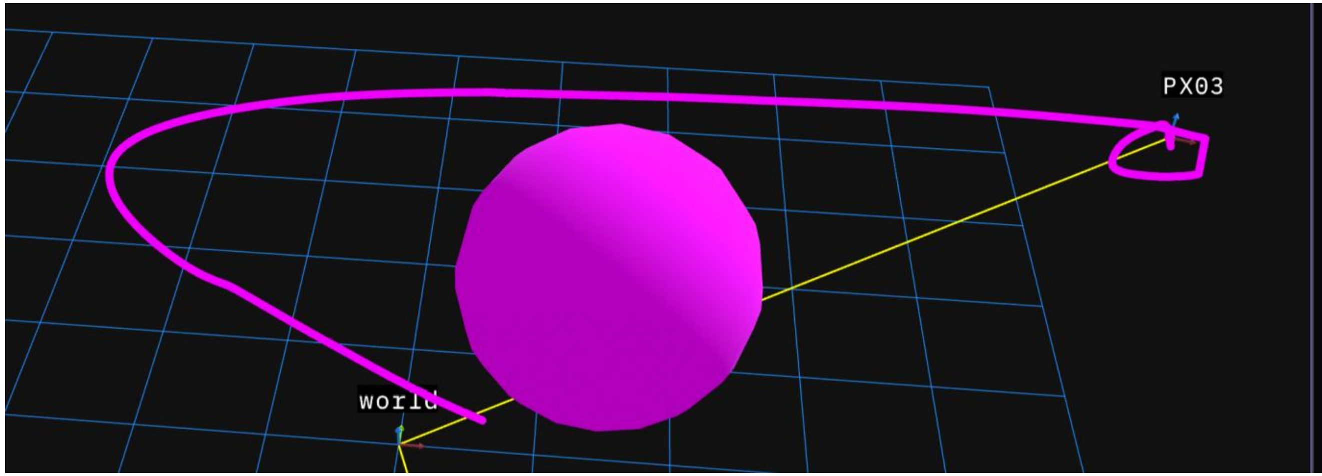
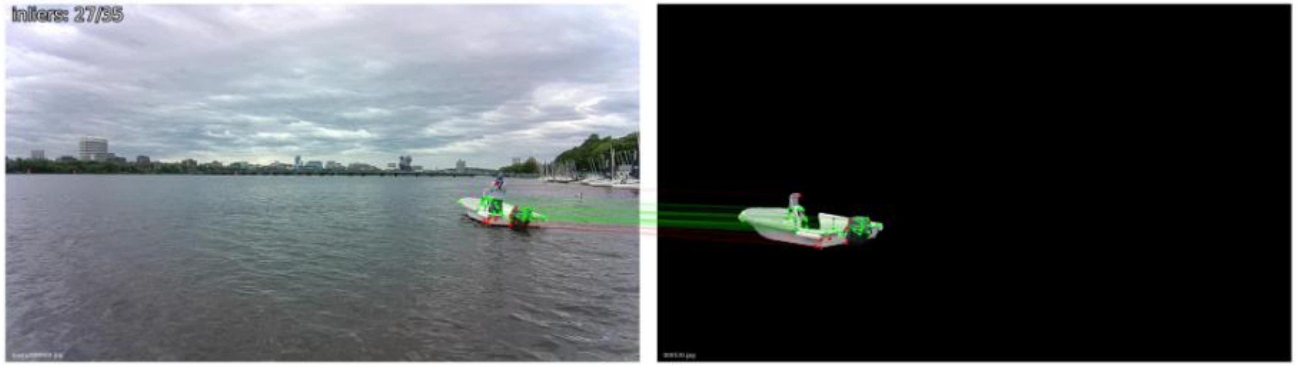
OBSTACLE AVOIDANCE ANALYTICAL CBF HARDWARE

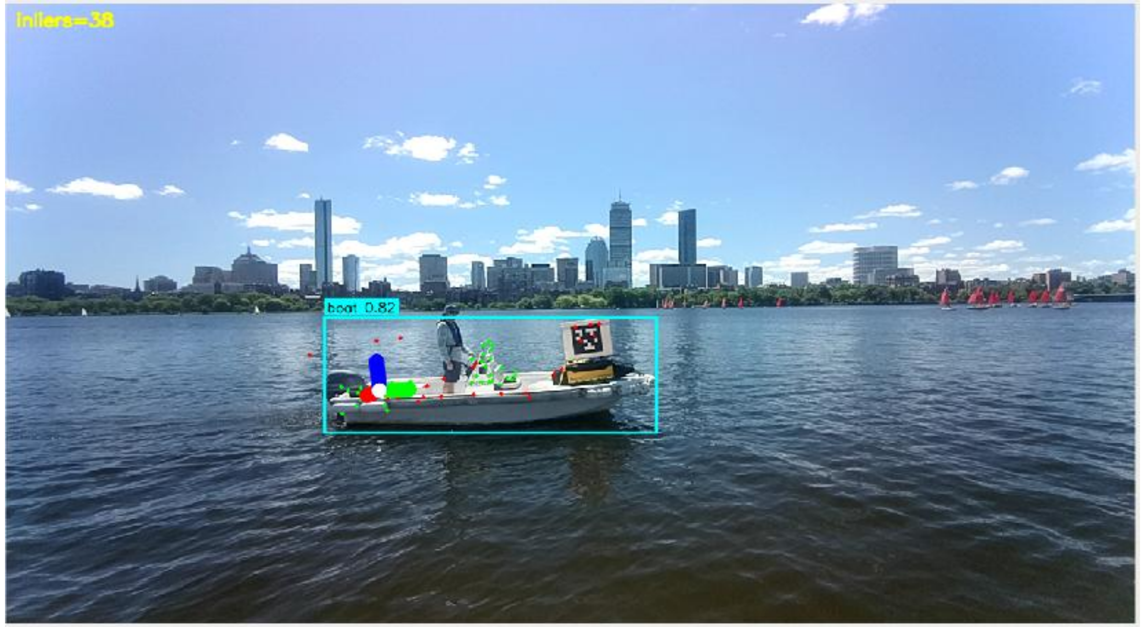
- Had issues with ROS2 namespaces
  - Goals and the CBF augmentation were not being received
- Algorithm could not be tested on the hardware
- Good learning experience
  - Need a better pipeline for HITL testing



HLOC STACK

- Helped improve the boat HLOC perception stack
  - Increased inlier count significantly
  - Improved the performance of HLOC pose estimate
  - Used an off the shelf YOLOV8-N Detector Model





LESSONS LEARNED

Technical

- **Hard**ware testing is hard
- Steep ramp-up on advanced controllers
- Visualization tools are essential

Professional

- First time with hard deadlines
- Fast-paced, experienced team

People & Culture

- In-office connections matter
- Great work culture boost productivity

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